

24/24

Excellent!

PHY202 Final Exam 0 on chapter(s): 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28

Name: _____

100%

These questions assess your understanding of the material from Chapter(s) 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units. Correct numerical answers do not guarantee credit.

1. Suppose that you rub a glass rod with silk, placing a 97.-nC positive charge on it. Calculate the initial force on the rod due to the Earth's magnetic field, if you throw the rod with a horizontal velocity of $5.8 \frac{m}{s}$ due west in a place where Earth's field is due north. Recall that the magnitude of Earth's magnetic field is $60. \mu T$.

$$q = 97 \times 10^{-9} C$$

$$v = 5.8 m/s$$

$$B = 60 \times 10^{-6} T$$

$$F = q v b \sin \theta$$

$$= (97 \times 10^{-9})(5.8)(60 \times 10^{-6})$$

$$= 3.38 \times 10^{-11} N$$

ANSWER: $3.38 \times 10^{-11} N$ ✓

2. The hot and neutral wires supplying DC power to a light-rail commuter train carry 900 A and are separated by 58. cm. What is the magnitude of the force between 250 m of these wires?

$$I = 900 A$$

$$r = 58 \times 10^{-2} m$$

$$l = 250 m$$

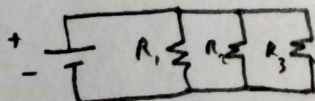
$$\frac{F}{l} = \frac{\mu_0 (I)^2}{2\pi r}$$

$$F = \frac{\mu_0 (900)^2}{2\pi (58 \times 10^{-2})} \times 250 = 69.83 N$$

ANSWER: $69.83 N$ ✓

3. Suppose the voltage supplied to a circuit is 8.8 V. There are three resistors wired in parallel: $R_1 = 2.3 \Omega$, $R_2 = 2.8 \Omega$, and $R_3 = 4.3 \Omega$. Calculate the power output of the battery.

ANSWER: $79.34 W$ ✓



$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\Rightarrow R_{eq} = 0.976 \Omega$$

$$P = \frac{V^2}{R} = \frac{(8.8)^2}{0.976} = 79.34 W$$

+3

14

Suppose an astronaut is traveling at 1.3×10^8 m/s from Earth to Alpha Centauri, which is a star system 4.0 ly away as measured by an Earth-bound observer. How far apart are Earth and Alpha Centauri as measured by the astronaut? Note: ly is lightyear, the distance light travels in one year.

$$L_0 = 4.0 \text{ ly}$$

$$\gamma = \left(\sqrt{1 - \frac{v^2}{c^2}} \right)^{-1}$$

$$= \left(\sqrt{1 - \frac{(1.3 \times 10^8)^2}{(3.00 \times 10^8)^2}} \right)^{-1}$$

$$= 1.11$$

$$L = L_0 \gamma^{-1}$$

$$= (4.0)(1.11)^{-1}$$

$$= (4.0)(0.90) = 3.60 \text{ ly}$$

ANSWER: 3.60 ly

(correct)

5. Calculate the power dissipated by a $3.7\text{-}\Omega$ car headlight when it has a voltage applied to it of 12. V.

ANSWER: 38.92 W

$$R = 3.7 \text{ }\Omega$$

$$V = 12 \text{ V}$$

$$P = \frac{V^2}{R} = \frac{(12)^2}{3.7} = 38.92 \text{ W}$$

6. Calculate the magnitude of the force exerted on a 4.7-nC charged particle placed in a $6.6 \times 10^4 \frac{\text{N}}{\text{C}}$ electric field.

ANSWER: $3.102 \times 10^{-4} \text{ N}$

$$E = 6.6 \times 10^4 \text{ N/C}$$

$$q = 4.7 \times 10^{-9} \text{ C}$$

$$F = E \cdot q$$

$$= (6.6 \times 10^4)(4.7 \times 10^{-9})$$

$$= 3.102 \times 10^{-4} \text{ N}$$

+2

7. Dry air will support electric fields up to its dielectric strength, $3.0 \times 10^6 \frac{\text{V}}{\text{m}}$. Above that value, an electric field creates enough ionization in the air to make the air a conductor, which causes a discharge or spark to reduce the field. Calculate the total electrostatic energy stored between a charged region at the base of a thundercloud and the ground if they are separated by 0.85 km of dry air. Model the cloud and ground as two parallel conducting plates and assume a thundercloud diameter of 4.0 km.

$$E = 3.0 \times 10^6 \text{ V/m} \quad r = 2000 \text{ m}$$

$$\begin{aligned} \Delta V &= E \cdot d \\ &= (3 \times 10^6) (0.85 \times 1000) \\ &= 2.55 \times 10^9 \text{ V} \end{aligned}$$

$$\begin{aligned} C &= \frac{\epsilon_0 A}{d} = \frac{\epsilon_0 (\pi r)^2}{d} \\ &= \frac{(8.85 \times 10^{-12}) (\pi) (2000)^2}{850} = 1.31 \times 10^{-9} \text{ F} \end{aligned}$$

ANSWER: $4.26 \times 10^9 \text{ J}$

$$\begin{aligned} E_{\text{cap}} &= \frac{CV^2}{2} \left(\times 10^{11} \right) \\ &= \frac{(1.31 \times 10^{-9}) (2.55 \times 10^9)^2}{2} \\ &= 4.26 \times 10^9 \text{ J} \end{aligned}$$

8. If the shock absorbers in a car go bad, then the car will oscillate at the smallest provocation, such as when going over bumps in the road and after stopping. Calculate the period of oscillations for a car with shoddy shock absorbers if the car's total mass is $2.53 \times 10^3 \text{ kg}$ and the force constant of the suspension system is $2.47 \times 10^4 \frac{\text{N}}{\text{m}}$.

$$\begin{aligned} T_{\text{SHO}} &= 2\pi \sqrt{\frac{m}{k}} \\ &= 2\pi \sqrt{\frac{2.53 \times 10^3}{2.47 \times 10^4}} \\ &= 2.01 \text{ s} \end{aligned}$$

ANSWER: 2.01 s

9. What angle would the axis of a polarizing filter need to make with the direction of polarized light of intensity 1.3 kW/m^2 to reduce the intensity to 0.66 kW/m^2 ?

$$I = I_0 \cos^2 \theta$$

$$\sqrt{\left(\frac{I}{I_0}\right)} = \cos \theta$$

$$\theta = \cos^{-1} \sqrt{\frac{I}{I_0}}$$

$$= \cos^{-1} \sqrt{\frac{(0.66 \times 10^3)}{(1.3 \times 10^3)}} = 44.56^\circ$$

ANSWER: 44.56°

+2

- ✓ 10. An electron moves at $3.2 \times 10^6 \frac{m}{s}$ perpendicular to a magnetic field. The field causes the electron to travel in a circular path of radius 18. cm. What is the strength of the magnetic field?

ANSWER: $1.01 \times 10^{-4} T$

$$v = 3.2 \times 10^6 \text{ m/s}$$

$$r = 18 \times 10^{-2} \text{ m}$$

$$r = \frac{mv}{qB}$$

$$\hookrightarrow B = \frac{mv}{qr}$$

$$= \frac{(9.11 \times 10^{-31})(3.2 \times 10^6)}{(1.60 \times 10^{-19})(18 \times 10^{-2})} = 1.01 \times 10^{-4} T$$

- ✓ 11. A scientist at a natural-history museum is holding a specimen 6.8 cm from a thin convex magnifying glass of focal length 8.4 cm. What magnification is produced?

ANSWER: 5.25

$$d_o = 6.8 \times 10^{-2} \text{ m}$$

$$f = 8.4 \times 10^{-2} \text{ m}$$

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

$$\frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o}$$

$$= \frac{1}{8.4 \times 10^{-2}} - \frac{1}{6.8 \times 10^{-2}}$$

$$= -2.80 \rightarrow d_i = -0.357 \text{ m}$$

$$m = \frac{-d_i}{d_o}$$

$$= \frac{-(-0.357)}{6.8 \times 10^{-2}}$$

$$= 5.25$$

- ✓ 12. What current flows through a high-voltage transmission line to transmit 65. MW of power at 420 kV?

ANSWER: 154.76 A

$$P = 65 \times 10^6 \text{ W}$$

$$V = 420 \times 10^3 \text{ V}$$

$$P = IV$$

$$I = \frac{P}{V}$$

$$= \frac{(65 \times 10^6)}{(420 \times 10^3)} = 154.76 A$$

+3

13. A sonar echo returns to a submarine 0.964 s after being emitted from the submarine. What is the distance from the submarine to the object, given that the speed of sound in this water is $1.51 \frac{\text{km}}{\text{s}}$?

$$t = \frac{0.964 \text{ s}}{2} = 0.482 \text{ s}$$

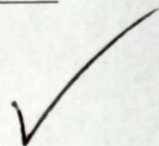
ANSWER: 727.82 m

$$v = \frac{x}{t}$$

$$x = v \cdot t$$

$$= (1.51 \times 10^3) (0.482)$$

$$= 727.82 \text{ m}$$



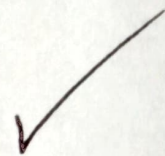
14. A medical imaging device produces ultrasound by oscillating with a period of $0.11 \mu\text{s}$. What is the frequency of this oscillation?

ANSWER: $9.09 \times 10^6 \text{ Hz}$

$$f = \frac{1}{T}$$

$$= \frac{1}{0.11 \times 10^{-6}}$$

$$= 9.09 \times 10^6 \text{ Hz}$$



3

15. On its highest power setting, a 450-W microwave oven projects microwaves onto a 0.24-m-by-0.24-m area. Calculate the peak electric-field strength in the electromagnetic waves emitted by the microwave oven.

ANSWER: 2.43 kV/m

$$P = 450 \text{ W}$$

$$A = (0.24)^2$$

$$I = \frac{P}{A} = \frac{450}{(0.24)^2} = 7812.5 \text{ W/m}^2$$

$$I = \frac{c \epsilon_0 E_0^2}{2}$$

$$\sqrt{\frac{2I}{c \epsilon_0}} = E_0 \rightarrow E_0 = 2.43 \text{ kV/m}$$

+2

- ✓ 16. A generator coil is rotated 0.55 revolutions in 6.3 ms. The 270-turn circular coil has a 2.1-cm radius and is in a uniform 0.76-T magnetic field. What is the maximum emf that is induced?

ANSWER: 155.94 V

$$\begin{aligned}\epsilon &= NAB\omega \sin(\omega t) \\ &= (270)(\pi)(2.1 \times 10^{-2})^2 (0.76)(549.53) \\ &= 155.94 \text{ V}\end{aligned}$$

$$\begin{aligned}\omega &= \frac{\Delta\theta}{\Delta t} = \frac{3.46}{6.3 \times 10^{-3}} \\ &= 549.53 \text{ rad/s}\end{aligned}$$

$$\theta = 0.55 \text{ rev} \left| \frac{2\pi}{1 \text{ rev}} \right. = 3.46 \text{ rad}$$

- ✓ 17. You shine a beam of white light through a diffraction grating with 1.9×10^4 lines per centimeter to a screen 1.4 m away. Find the angle for first-order diffraction of 490 nm light.

ANSWER: 68.59°

$$N = 1.9 \times 10^4 \frac{\text{lines}}{\text{cm}} \left| \frac{100 \text{ cm}}{1 \text{ m}} \right. = 1900 \times 10^3 \text{ m}$$

$$d = \frac{1}{N} = 5.26 \times 10^{-7} \text{ m}$$

$$d \sin \theta = m \lambda$$

$$\theta = \sin^{-1} \left(\frac{m \lambda}{d} \right)$$

$$= \sin^{-1} \left(\frac{(1)(490 \times 10^{-9})}{5.26 \times 10^{-7}} \right) = 68.68^\circ$$

- ✓ 18. Calculate the sound intensity level in decibels for a sound wave traveling in air at 3.8°C with a pressure amplitude of 4.98×10^{-3} Pa. Recall that the density of air is $1.29 \frac{\text{kg}}{\text{m}^3}$ and the speed of sound in air at 0°C is $331 \frac{\text{m}}{\text{s}}$.

ANSWER: 44.60 dB

$$\begin{aligned}I &= \frac{(\Delta P)^2}{2 \rho v_w} \\ &= \frac{(4.98 \times 10^{-3})^2}{2(1.29)(333.30)} \\ &= 2.88 \times 10^{-9} \text{ A}\end{aligned} \quad \left| \begin{aligned}v_w &= (331) \sqrt{\frac{T}{273}} \\ &= (331) \sqrt{\frac{276.8}{273}} \\ &= 333.30 \text{ m/s}\end{aligned} \right.$$

$$B = 10 \log \left(\frac{I}{I_0} \right)$$

$$+ 3 = 44.60 \text{ dB}$$

19. Find the maximum torque on a 140-turn square loop of wire that is 4.0 cm on each side that carries 1.5 A of current in a 0.49-T field.

ANSWER: 0.16 N·m

$$\begin{aligned} \tau &= N I A B \sin \theta \\ &= (140)(1.5)(4 \times 10^{-2})^2 (0.49) \\ &= 0.16964 \end{aligned}$$

✓

20. Calculate the magnitude of the electrostatic force between an electron and a proton separated by 2.94 pm.

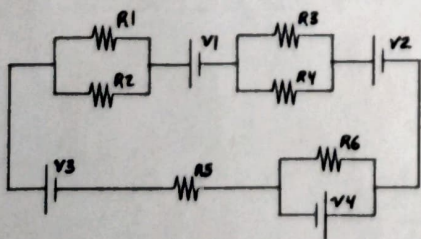
ANSWER: $2.66 \times 10^{-5} \text{ N}$

$$\begin{aligned} F &= \frac{k |q_1 q_2|}{r^2} \\ &= \frac{(8.99 \times 10^9) (1.60 \times 10^{-19})^2}{(2.94 \times 10^{-12})^2} \\ &= 2.66 \times 10^{-5} \text{ N} \end{aligned}$$

$(2.66 \times 10^{-5} \text{ N})$ ✓

- X 21. Solve for the power dissipated in resistor R_3 given $V_1 = 8 \text{ V}$, $V_2 = 10 \text{ V}$, $V_3 = 4 \text{ V}$, $V_4 = 4 \text{ V}$, $R_1 = 1 \Omega$, $R_2 = 1 \Omega$, $R_3 = 2 \Omega$, $R_4 = 1 \Omega$, $R_5 = 2 \Omega$, and $R_6 = 2 \Omega$.

ANSWER: _____



4

72

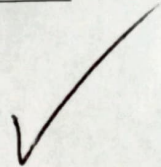
22. What is the maximum strength of the magnetic field in an electromagnetic wave that has a maximum electric-field strength of 270 V/m?

ANSWER: $9.00 \times 10^{-7} \text{ T}$

$$\frac{\epsilon_0 E_0^2}{2} = \frac{\mu_0 B_0^2}{2}$$

$$(8.85 \times 10^{-12})(270)^2 = \frac{B_0^2}{4\pi \times 10^{-7}}$$

$$B_0 = 9.00 \times 10^{-7} \text{ T}$$



23. A clear-glass light bulb is placed 0.96 m from a thin convex lens that has a focal length of 0.51 m. Calculate the magnification of the produced image.

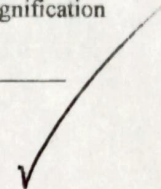
ANSWER: -1.13

$$d_o = 0.96 \text{ m}$$

$$f = 0.51 \text{ m}$$

$$\frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o} = \frac{1}{0.51} - \frac{1}{0.96} \Rightarrow d_i = 1.088 \text{ m}$$

$$m = -\frac{d_i}{d_o}$$



24. Calculate the self-inductance of a 0.37 m-long, 0.014 m-diameter solenoid that has 230 coils.

ANSWER: $2.77 \times 10^{-5} \text{ H}$

$$L = \frac{\mu_0 N^2 A}{l}$$

$$= \frac{(4\pi \times 10^{-7})(230)^2 (\pi)(7 \times 10^{-3})^2}{0.37}$$

$$= 2.77 \times 10^{-5} \text{ H}$$



+3

25. Suppose you have a tube that is closed at one end. Its length is 13. cm and the air temperature is 2.3°C. What is the frequency of its third overtone?

ANSWER: 4474.50 Hz ✓

$$f_n = \frac{n v_w}{4L}$$

$$= \frac{7 (332.39)}{4 (13 \times 10^{-2})}$$

$$= 4474.50 \text{ Hz}$$

$$v_w = (331) \sqrt{\frac{2.3 + 273}{273}}$$

$$= 332.39 \text{ m/s}$$

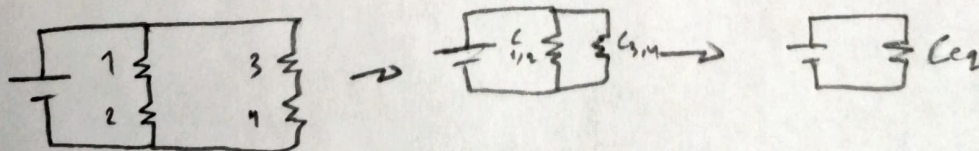
- X 26. Suppose a galaxy is moving away from Earth at a speed of 2.4×10^8 m/s. It emits radio waves with a wavelength of 0.21 m. What wavelength would we detect on Earth?

ANSWER: _____

5

27. A circuit has two branches: one branch has capacitors 1 and 2 while the second branch has capacitors 3 and 4. Capacitors 1 and 2 are connected in series and capacitors 3 and 4 are connected in series. The two branches are connected in parallel. Draw this circuit and find the total capacitance given that $C_1 = 1.3 \text{ nF}$, $C_2 = 23. \text{ nF}$, $C_3 = 1.3 \text{ nF}$, and $C_4 = 1.2 \text{ nF}$. Note: you must draw the circuit correctly to receive credit.

ANSWER: $1.854 \times 10^{-9} \text{ F}$ ✓



$$\frac{1}{C_{1,2}} = \frac{1}{1.3 \text{ nF}} + \frac{1}{23 \text{ nF}} \Rightarrow C_{1,2} = 1.23 \times 10^{-9} \text{ F}$$

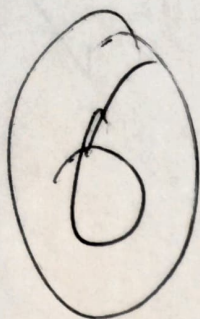
$$\frac{1}{C_{3,4}} = \frac{1}{1.3 \text{ nF}} + \frac{1}{1.2 \text{ nF}} \Rightarrow C_{3,4} = 6.24 \times 10^{-10} \text{ F}$$

$$C_{eq} = C_{1,2} + C_{3,4} = 1.854 \times 10^{-9} \text{ F}$$

12

- X 28. What is the size of the image on the retina of a 2.8 mm-diameter strand of hair that is held at arm's length (53. cm) away? Take the lens-to-retina distance to be 1.6 cm.

ANSWER: _____



- ✓ 29. What is the magnitude of the maximum force exerted on an aluminum rod with a 1.0-μC charge that you pass between the poles of a 0.17-T permanent magnet at a speed of 9.5 $\frac{m}{s}$?

ANSWER: $1.615 \times 10^{-6} \text{ N}$

$$\begin{aligned} F &= q v b \sin \theta \\ &= (1.0 \times 10^{-6})(9.5)(0.17) \\ &= 1.615 \times 10^{-6} \text{ N} \end{aligned}$$

- ✓ 30. Suppose that you take a magnifying glass out on a sunny day and you find that it concentrates sunlight to a small spot 1.4 cm away from the lens. What is the power of the lens?

ANSWER: 71.43 D

$$\begin{aligned} P &= \frac{1}{f} \\ &= \frac{1}{1.4 \times 10^{-2}} \\ &= 71.43 \text{ D} \end{aligned}$$

x2

PHY202 Final Exam on chapter(s): 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28

1. $F = 3.38 \times 10^{-11} \text{ N} \therefore F = qvbsin\theta, \theta = 90^\circ$
(Chapter 22, Section (4) Magnetic Field Strength - cple_ch22_ex1)
2. $F = 69.8 \text{ N} \therefore \frac{F}{\ell} = \frac{\mu_0 I_1 I_2}{2\pi r}$
(Chapter 22, Section (10) Magnetic Force between Two Parallel Conductors - cple_ch22_p50)
3. $P = 79.3 \text{ W} \therefore \frac{1}{R_p} = \frac{1}{R_1} + \dots + \frac{1}{R_n}, V = IR, P = IV$
(Chapter 21, Section (1) Resistors in Series and Parallel - cple_ch21_ex2)
4. $L = 3.60 \text{ ly} \therefore L = L_0 \gamma^{-1}$
(Chapter 28, Section (3) Length Contraction - cple_ch28_ex2)
5. $P = 38.9 \text{ W} \therefore P = \frac{V^2}{R}$
(Chapter 20, Section (4) Electric Power and Energy - cple_ch20_ex7)
6. $F = 3.10 \times 10^{-4} \text{ N} \therefore F = Eq$
(Chapter 18, Section (4) Electric Field - cple_ch18_ex3)
7. $E_{\text{stored}} = 4.25 \times 10^{11} \text{ J} \therefore C = \epsilon_0 \frac{A}{d}, \Delta V = Ed, E_{\text{cap}} = \frac{CV^2}{2}$
(Chapter 19, Section (2) Electric Potential in a Uniform Electric Field - cple_ch19_ex4)
8. $f = 2.01 \text{ s} \therefore T_{\text{SHO}} = 2\pi \sqrt{\frac{m}{k}}$
(Chapter 16, Section (3) Simple Harmonic Motion - cple_ch16_ex4)
9. $\theta = 44.6^\circ \therefore I = I_0 \cos^2 \theta$
(Chapter 27, Section (8) Polarization - cple_ch27_p87)
10. $B = 1.01 \times 10^{-4} \text{ T} \therefore r = \frac{mv}{qB}, m_e = 9.11 \times 10^{-31} \text{ kg}, q_e = 1.60 \times 10^{-19} \text{ C}$
(Chapter 22, Section (5) Force on a Moving Charge in a Magnetic Field - cple_ch22_p13)
11. $m = 5.25 \therefore \frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}, \frac{h_i}{h_o} = -\frac{d_i}{d_o} = m$
(Chapter 26, Section (6) Image Formation by Lenses - cple_ch25_ex7)
12. $I_{\text{RMS}} = 155. \text{ A} \therefore P_{\text{ave}} = I_{\text{RMS}} V_{\text{RMS}}$
(Chapter 20, Section (5) Alternating Current v. Direct Current - cple_ch20_ex10)
13. $d = 728. \text{ m} \therefore \vec{v}_{\text{avg}} = \frac{\Delta \vec{x}}{\Delta t}$
(Chapter 17, Section (2) Speed of Sound, Frequency, Wavelength - cple_ch17_p8)
14. $f = 9.09 \times 10^6 \text{ Hz} \therefore T = \frac{1}{f}$
(Chapter 16, Section (2) Period and Frequency in Oscillations - cple_ch16_ex3)
15. $E_0 = 2.43 \times 10^3 \frac{\text{V}}{\text{m}} \therefore I_{\text{ave}} = \frac{c\epsilon_0 E_0^2}{2} = \frac{cB_0^2}{2\mu_0} = \frac{E_0 B_0}{2\mu_0}$
(Chapter 24, Section (4) Energy in Electromagnetic Waves - cple_ch24_ex4)
16. $\epsilon = 156. \text{ V} \therefore \epsilon = NAB\omega \sin(\omega t), \sin(\omega t) = 1$
(Chapter 23, Section (5) Electric Generators - cple_ch23_ex4)

17. $\theta = 68.6^\circ \therefore d \sin \theta = m \lambda$
(Chapter 27, Section (4) Multiple-Slit Diffraction - cp1e_ch27_ex3)
18. $\beta = 44.6 \text{ dB} \therefore v_w = \left(331 \frac{\text{m}}{\text{s}}\right) \sqrt{\frac{T}{273 \text{ K}}}$, $I = \frac{(\Delta p)^2}{2 \rho v_w}$, $\beta \text{ (dB)} = 10 \log_{10} \left(\frac{I}{I_0}\right)$, $I_0 = 1 \times 10^{-12} \frac{\text{W}}{\text{m}^2}$
(Chapter 17, Section (3) Sound Intensity and Sound Level - cp1e_ch17_ex2)
19. $\tau = 0.165 \text{ N} \cdot \text{m} \therefore \tau = NIAB \sin \theta \therefore \tau_{\max} = NIAB$
(Chapter 22, Section (8) Torque on a Current Loop: Motors and Meters - cp1e_ch22_ex5)
20. $F = 2.66 \times 10^{-5} \text{ N} \therefore F = k \frac{|q_1 q_2|}{r^2}$
(Chapter 18, Section (3) Coulomb's Law - cp1e_ch18_ex1)
21. $P = 7.18 \text{ W} \therefore$ Kirchhoff's rules and $P = I^2 R$
(Chapter 21, Section (3) Kirchhoff's Rules - tkd_circ2)
22. $B_{\max} = 9.00 \times 10^{-7} \text{ T} \therefore c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = \frac{E}{B} = f \lambda$
(Chapter 24, Section (2) Production of Electromagnetic Waves - cp1e_ch24_ex1)
23. $m = -1.13 \therefore \frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}$, $\frac{h_i}{h_o} = -\frac{d_i}{d_o} = m$
(Chapter 25, Section (6) Image Formation by Lenses - cp1e_ch25_ex6b)
24. $L = 2.77 \times 10^{-5} \text{ H} \therefore L = N \frac{\Delta \Phi}{\Delta i} = \frac{\mu_0 N^2 A}{\ell}$
(Chapter 23, Section (9) Inductance - cp1e_ch23_ex7)
25. $f = 4.47 \times 10^3 \text{ Hz} \therefore f_n = n \frac{v_w}{4L}$, $v_w = \left(331 \frac{\text{m}}{\text{s}}\right) \sqrt{\frac{T}{273 \text{ K}}}$
(Chapter 17, Section (5) Sound Interference and Resonance: Standing Waves in Air Columns - cp1e_ch17_ex5b)
26. $\lambda_{\text{obs}} = 0.630 \text{ m} \therefore \lambda_{\text{obs}} = \lambda_s \sqrt{\frac{1 + \frac{u}{c}}{1 - \frac{u}{c}}}$
(Chapter 28, Section (4) Relativistic Addition of Velocities - cp1e_ch28_ex5)
27. $C_T = 1.85 \times 10^{-9} \text{ F} \therefore \frac{1}{C_s} = \frac{1}{C_1} + \dots + \frac{1}{C_n}$, $C_p = C_1 + \dots + C_n$
(Chapter 19, Section (6) Capacitors in Series and Parallel - cp1e_ch19_ex10)
28. $h_i = -8.45 \times 10^{-5} \text{ m} \therefore \frac{h_i}{h_o} = -\frac{d_i}{d_o} = m$
(Chapter 26, Section (1) Physics of the Eye - cp1e_ch26_ex1)
29. $F = 1.61 \times 10^{-6} \text{ N} \therefore F = qvb \sin \theta \therefore F_{\max} = qvb$
(Chapter 22, Section (4) Magnetic Field Strength - cp1e_ch22_p7)
30. $P = 71.4 \text{ D} \therefore P = \frac{1}{f}$
(Chapter 25, Section (6) Image Formation by Lenses - cp1e_ch25_ex5)